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Variance in Basketball: Optimizing Shot Selection

Abstract

Conventional basketball wisdom often prioritizes scoring efficiency, maximizing expected value, and playing for low volatility, but this approach may be suboptimal for teams facing a talent disadvantage or scoring deficit. This paper analyzes the role of variance in shooting strategies, drawing parallels to “high-volatility” tactics in other sports (e.g., pulling the goalie in hockey). Using Monte Carlo simulations (n = 1,000,000) and binomial distribution analysis, we model two scenarios: teams trailing by 10 points in the fourth quarter, and playoff teams facing superior opponents (e.g., 8-seeds vs. 1-seeds in the NBA).

Our results demonstrate that increasing the proportion of three-point attempts significantly improves win probability in two distinct high-leverage scenarios: trailing by double digits in the fourth quarter, and facing a superior opponent in a playoff series. Specifically, we identify a 'tipping point' of 32.5% 3-point efficiency, above which high-variance strategies consistently outperform conservative play.

Introduction

A helpful way to think about variance in basketball is to look at how other sports handle moments when expected value is no longer the main objective. When a team is trailing and time is running out, the right strategy often becomes the one that increases volatility, even if it lowers average efficiency. Other sports make this shift very clearly.

In hockey, the best example is pulling the goalie. Removing the goalie obviously increases the risk of giving up an empty net goal, but the added attacker boosts scoring chances more than any regular tactical adjustment. Teams wait too long to do it, but the idea is simple: the only thing that matters is the chance of tying the game.

Soccer follows a similar logic. When a team is down late, the goalkeeper comes up for a corner. This move exposes the entire field to a counterattack, but it meaningfully increases the odds of finding a last chance goal. Teams also push extra players forward and reshape the formation to create more shot volume. It is a good example of throwing stability out the window to chase a single high impact outcome.

Baseball has its own version with intentional walks. Managers sometimes choose to load the bases or put a high-power hitter on first to reduce the chance of the big negative outcome and increase the volatility of the inning.

Football is full of explicit variance decisions. Teams go for a two point attempts more often when trailing behind late in the game, even if the expected points per attempt are lower than kicking the extra point. It gives the trailing team more paths to winning instead of simply catching up. Onside kicks work the same way. They are very low probability, but they generate extra possessions that the losing team needs to change the game state.

All of these situations have the same structure: when you are trailing by enough that playing safe guarantees a loss, increasing variance becomes the rational move. Basketball fits into that framework, but do basketball teams actually do this? This paper will attempt to analyze variance in three-point shooting in the NBA and how shooting a higher number of three-point attempts affects win probability.

We will look at two different scenarios:

1. Should underdog teams in the playoffs increase their ratio of three-point attempts to two-point attempts in the playoffs, when the stakes are highest? And do they?
2. Do teams switch strategy to attempt more three-point shots when trailing by large margins late in games?

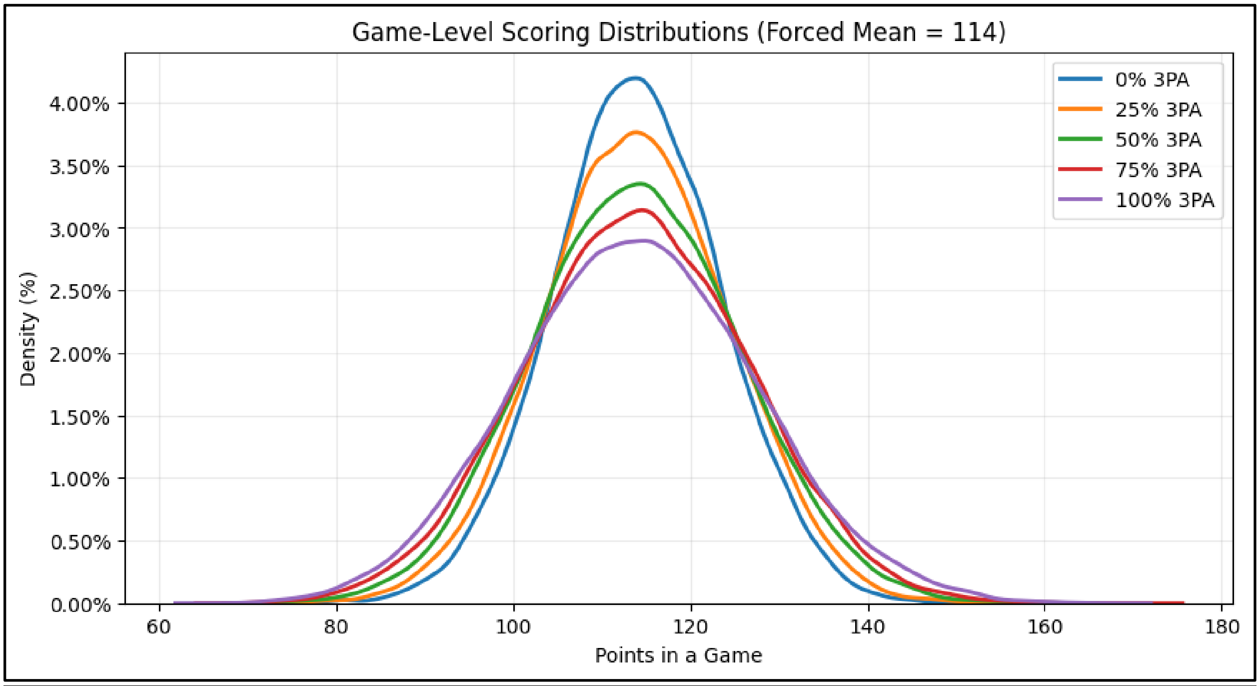
Background: The three-point revolution

For those who have followed basketball, the past decade has seen a revolution in how the game is played, with a significant increase in three-point shot attempts. At an investment forum, Daryl Morey, Philadelphia 76ers President of Basketball Operations, laid out the math clearly: $3 \times 0.4 > 2 \times 0.5$ (i.e., the number of points possible for the shot attempt multiplied by the probability of making the shot). The emergence of Stephen Curry's prolific three-point shooting and Morey's analytical approach fueled a meteoric rise in three-point attempts.

The numbers don't always multiply out that way when looking at expected points, since two-pointers closer to the basket (e.g., layups and dunks) are higher quality shots with higher expected values. In fact, on average, the expected points for both two-point and three-point shots tend to be similar. In reality, three-point make probability is roughly 36%, while two-point make probability is around 54%.

NBA teams attempt roughly 90 shots per game, so once we fix for total attempts (i.e., remove other factors like offensive rebounding and turnovers), the main driver of scoring volatility is the allocation between three-pointers and two-pointers. Because a three-point shot has a lower likelihood of success but a higher payoff, increasing the share of three-point attempts (3PA) inherently widens the scoring distribution. The mean stays anchored because overall efficiency doesn't change much, but the standard deviation increases as the point value of each shot becomes more variable. In the simulated distributions, this shows up as heavier tails and a flatter peak as 3PA in proportion to two-point attempts (2PA) rises. In other words, adjusting the shot mix alters the probability mass in a way that increases the likelihood of both very high and very low scoring outcomes. That is why attempting more three-pointers becomes strategically relevant when a team needs to access more "upside" (higher possible point swing) late in games.

Illustrative Example Showing scoring distribution

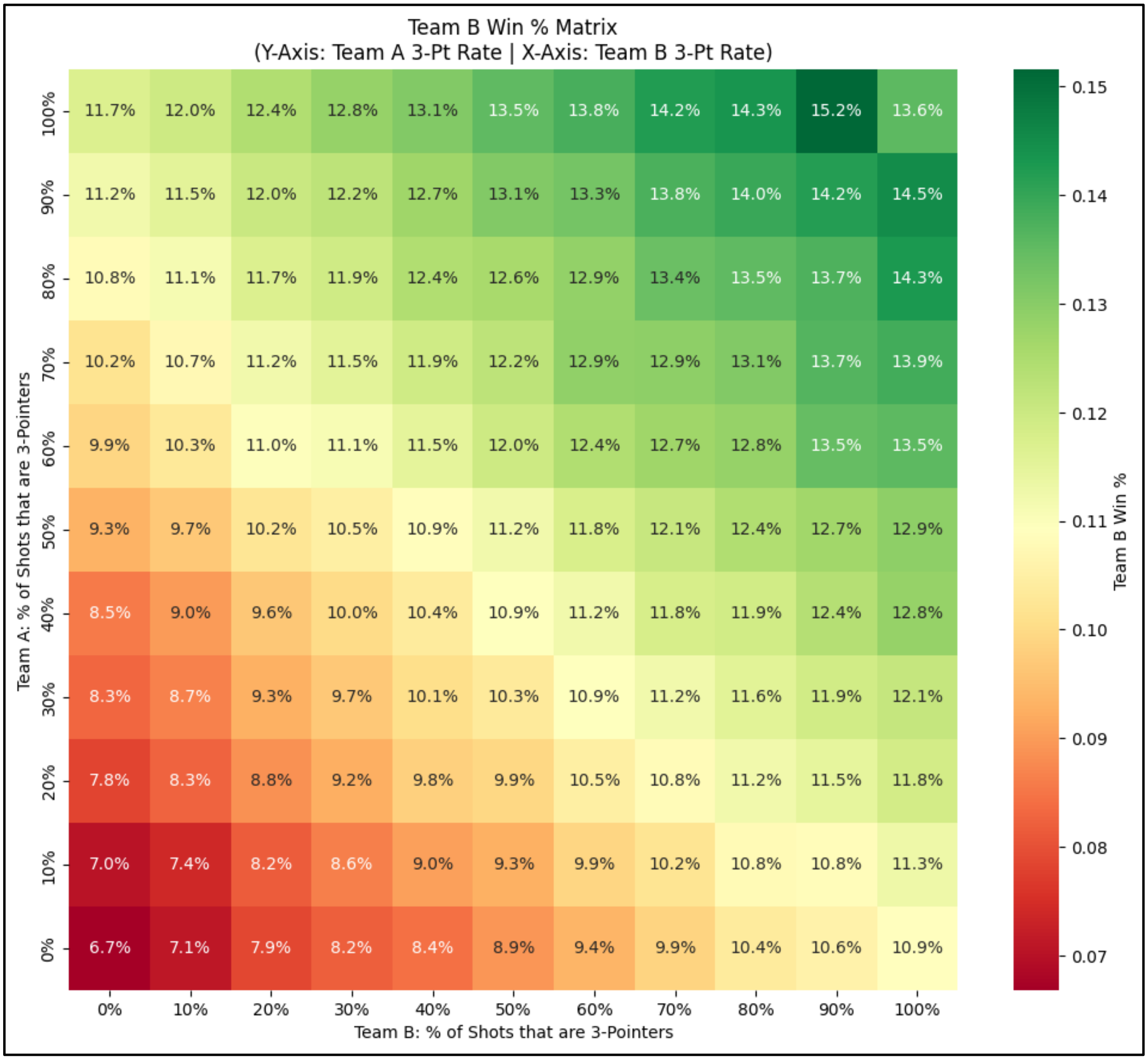


Preliminary Analysis

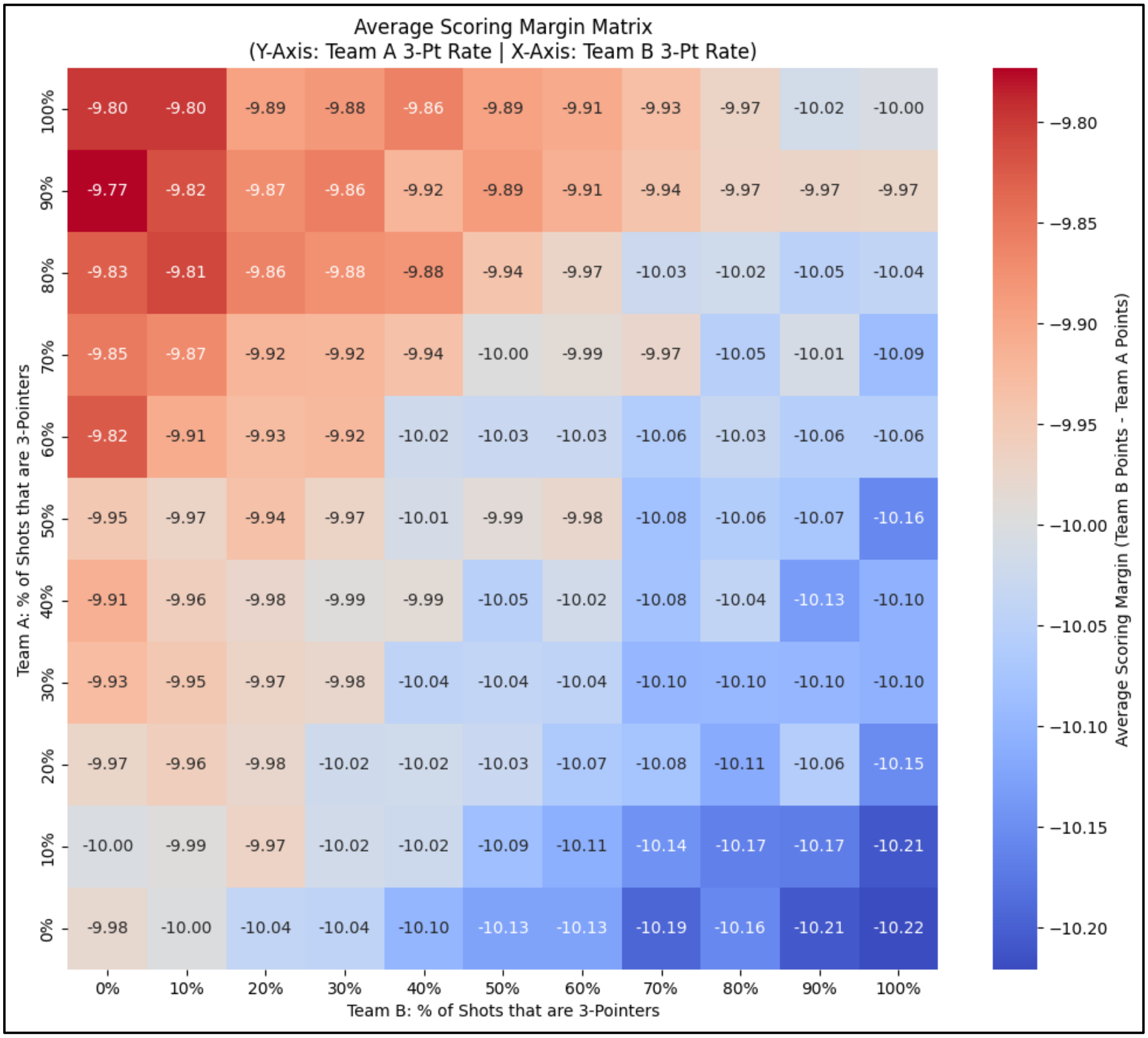
To quantify the impact of shooting variance on win probability, we started with a very simple model comparing outcomes for two evenly-matched teams, Team A and Team B, when Team B is trailing by 10 points at the start of the fourth quarter. We used aggregate shooting data from the 2024-25 season to run a Monte Carlo simulation, varying the proportion of three-point attempts in increments of 10 (0% of shot attempts are 3s versus 10%, 20%, etc). The parameters are as follows:

- Simulations per cell: 100,000
- Time remaining: 12 minutes
- Shot attempts per team per minute: 1.86
- Two-point make percentage: 54.5%
- Three-point make percentage: 36.0%

The model suggests that Team B should resort to almost exclusively shooting three-pointers to maximize their win percentage in situations like these, as suggested by the heatmap below:



Interestingly, as win probability rises with more three-point attempts, average margin of defeat actually *worsens***.** This is due to the additional variance caused by three-point shooting. Teams that are losing increase the likelihood of coming back to win, but simultaneously increase the likelihood that they lose by larger margins. The average point differential heatmap is visualized below:



While the gap in scoring margin between shooting no threes and attempting 100% of shot attempts as threes is only about 0.2 points, regardless of how Team A behaves, this is aggregated across 100k simulations for every cell. So while shooting a higher proportion of threes increases win probability, teams also experience blowout losses at a higher frequency. This is relevant because we know that coaches and teams have historically preferred to avoid the “negative optics” around losing by larger margins, contributing to why NFL teams have traditionally punted at disproportionately high rates on 4th down and why NHL teams avoid “pulling the goalie” and losing by larger margins.

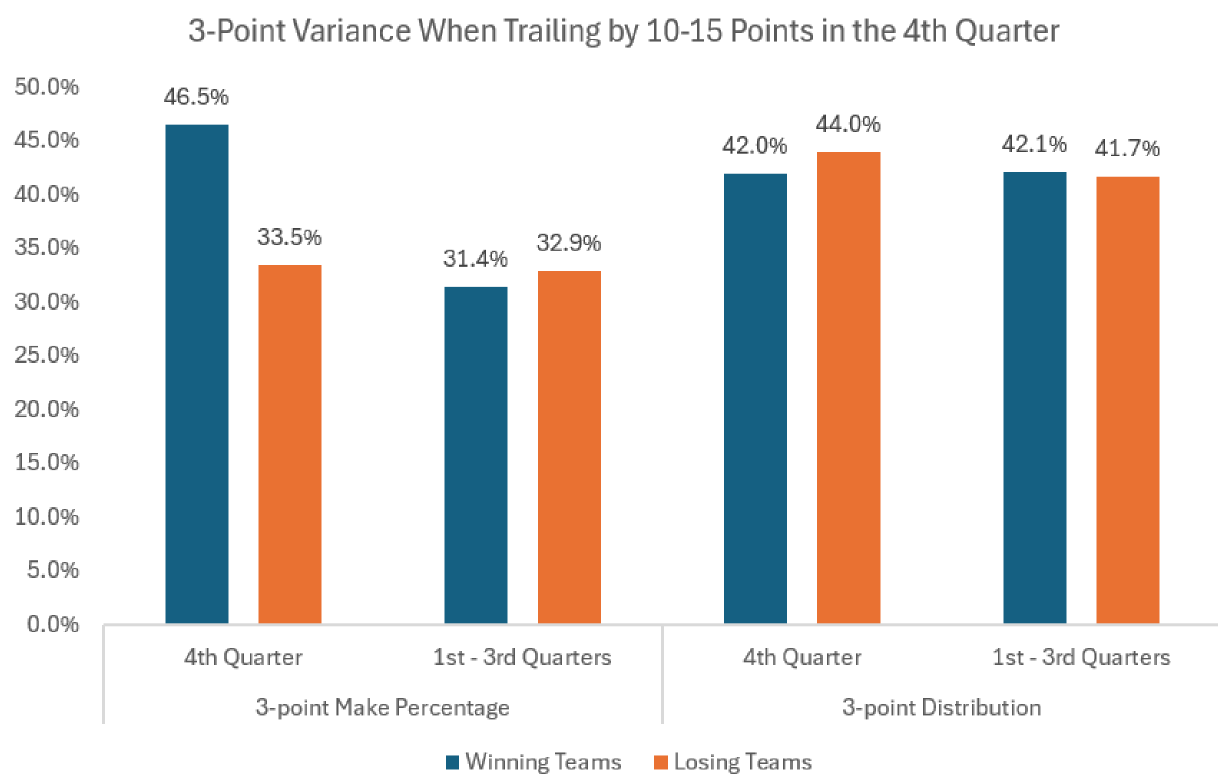
We hypothesize the same phenomena exists in the NBA, where teams losing late in games are reluctant to shoot larger shares of threes to avoid the risk of turning 10-point losses into 30-point losses (e.g., coaches fear losing their jobs because of blowout losses). In the next section, we analyze play-by-play data from the NBA to understand how frequently teams come back to win when trailing by large margins, how they change their two-point vs. three-point shot profile, and whether those outcomes align with what our model suggests.

Do NBA teams actually adjust their approach in late-game situations?

Our model estimates that teams are almost always better off shooting more threes to improve their chances of achieving a 'high standard deviation' outcome that leads to a win, so we looked at real play-by-play data from the 2024-25 NBA season to understand how teams behave when trailing in the 4th quarter of games.

There were 268 games in the 2024-25 regular season where a team trailed by 10-15 points entering the 4th quarter. The trailing team came back to win in 20 such instances, resulting in a 7.5% win percentage. What's interesting is that there were minimal differences in three-point shot distribution in the 4th quarter compared to the first three quarters of these games. The difference between the winning and losing teams came entirely from the winning teams making a disproportionately high number of their shot attempts.

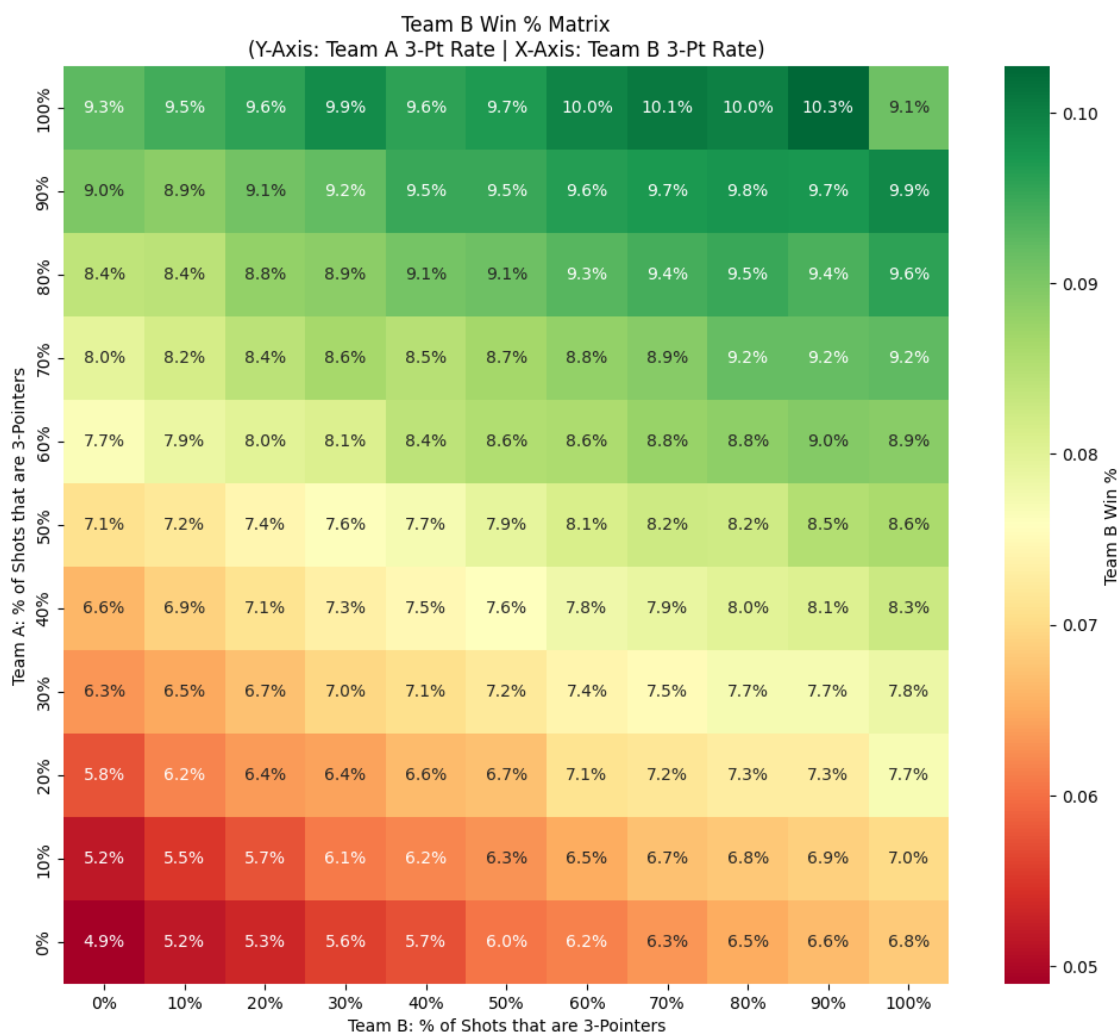
The chart below illustrates this distinction:



As we can see, winning teams *make* more threes, but do not necessarily *attempt* more threes. We wanted to understand if teams could improve upon the 7.5% win probability in situations like these, so we updated our Monte Carlo simulation using real-world shooting performance in these scenarios.

In the 248 instances that teams lost when trailing by 10-15 points at the start of the 4th quarter, they shot 52.1% on 2-point attempts and 32.9% on 3-point attempts in the first three quarters of these games. Note that these are both below league average, likely due to some combination of bad luck and the fact that teams that are losing in the 4th quarter are more likely to be bad teams. We updated our Monte Carlo simulation so that ‘Team B’ reflects the losing teams with these below-average shooting percentages, while holding ‘Team A’ at league-average efficiency.

The model suggests that even teams with below-average efficiency can improve their odds of winning from 0.5-1.1 percentage points by adjusting their shot selection from 40% three-pointers to resorting almost exclusively to shooting threes in the 4th quarter when losing, as visualized in the matrix below.

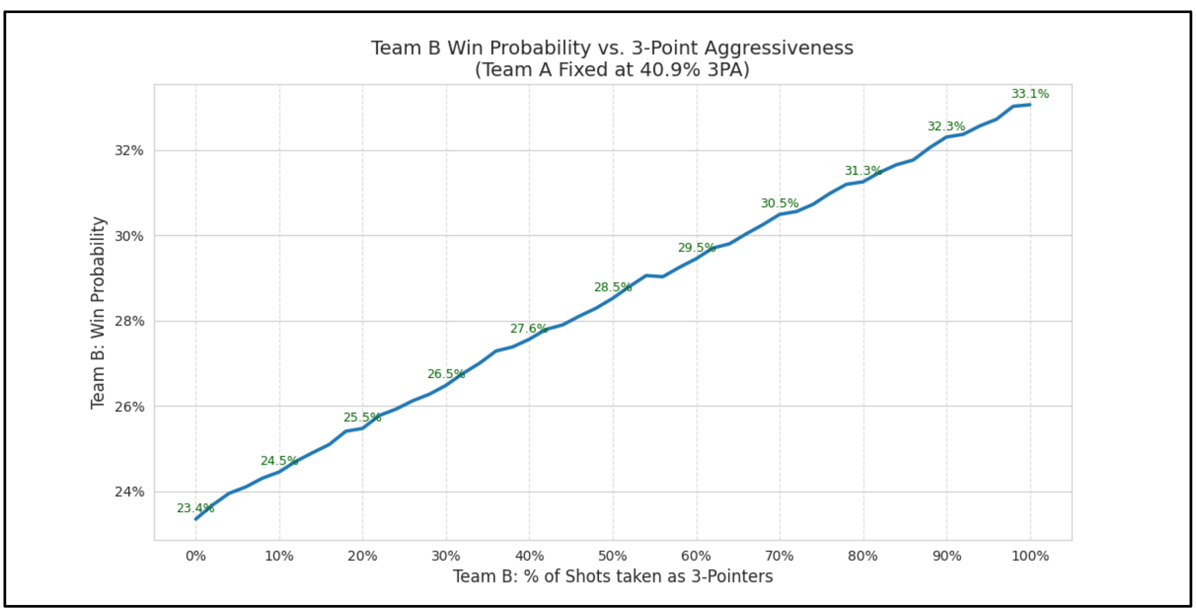


These results indicate that even teams that are faced with a ‘skill gap’ can benefit from attempting three-pointers at a higher frequency when trailing late in games. In the next section, we'll go in depth into how this comes into play when evaluating performances of underdogs in NBA playoff series.

Underdog analysis: Less-skilled teams aren't shooting enough threes

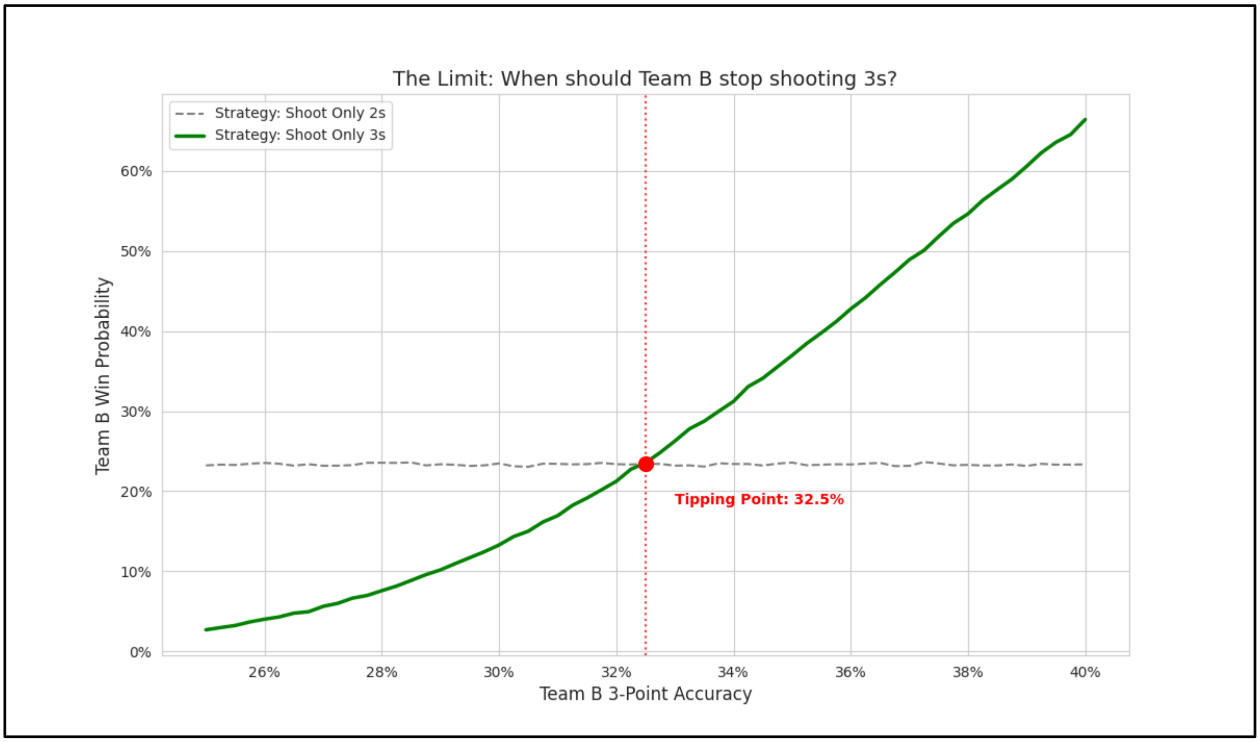
Another interesting insight emerges when we compare underdogs and favorites in the NBA postseason. From the variance logic above, we know underdogs might want to increase three-point volume to boost the likelihood of "getting lucky." However, underdog teams do not follow this logic in the playoffs. Over the last 10 years, eight-seeded teams took a lower proportion of three-pointers than one-seeded teams (34.7% vs. 40.9%), perhaps because they shoot worse from three (34.0% vs. 37.9%) See Exhibit A in the appendix for additional make percentages.

This raises the question: given the differences in 3pt shooting percentages, *should* lower seeded teams shoot as many threes as higher seeded teams? In a Monte Carlo simulation (n=1,000,000), we observe that shooting more threes **increases win probability for lower-seeded teams** (called "Team B" in the figure) *even though lower seeded teams have lower three-point success rates than higher seeded teams.* The figure below demonstrates how a lower seeded team's win percentage increases as it shoots a higher share of threes. For example, if the lower seeded team increases its share of 3pt shots from 30% to 60%, its win probability increases by 3 percentage points assuming its 2pt and 3pt make percentages remain the same. In fact, taking *only* threes theoretically maximizes the lower seed's win probability, though this is likely an unrealistic strategy in practice, as the opponent will crowd the 3pt line and 3pt make percentage will decrease.

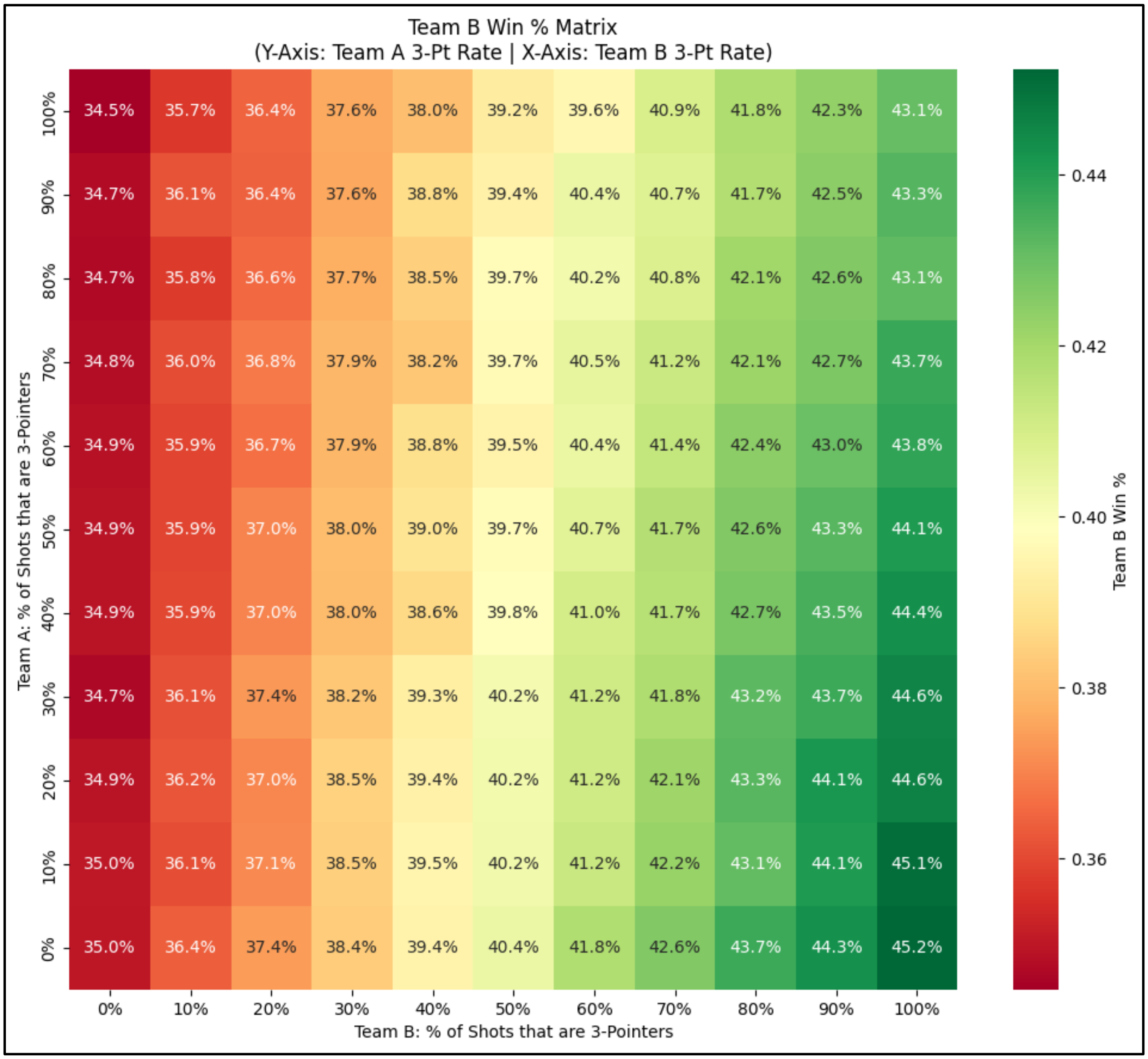


The takeaway is that eight-seeded teams should shoot more threes assuming 2pt and 3pt make percentages remain the same. However, there is a 3pt make percentage lower threshold–32.5%–

at which eight-seeded team should prioritize 2pt shots over 3pt shots. At 3pt make percentages lower than this, the expected value of 3pt shots is so low that teams gain no advantage via additional variance. The figure below demonstrates this.



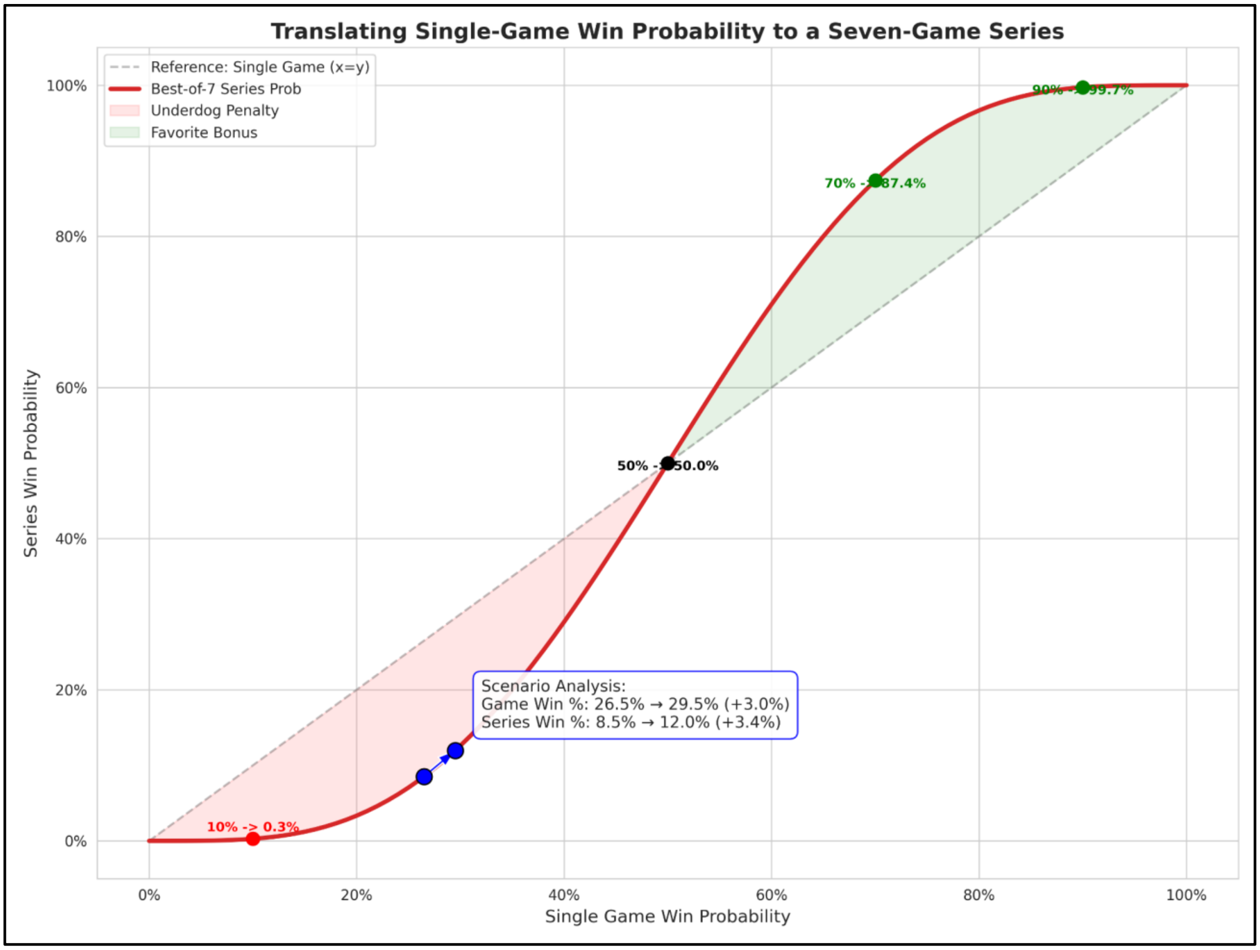
But what if the one-seeded team adjusts its share of 3pt shots in response to the eight-seeded team taking many threes? A Monte Carlo simulation demonstrates that *it doesn't matter.* The eight-seed is always better off taking more threes *regardless* of how the one-seeded team shoots. The heat map below demonstrates this.



Translating single-game win probability to series win probability

The 3 percentage point increase in per-game win percentage may seem small, but it translates to a 4.5 percentage point increase (8.0% to 12.5%) probability of winning the series. Over 20 years, this increases the likelihood of an underdog team winning at least one series from ~81% to ~93%. For teams that often find themselves in the underdog position, this is significant. See the first exhibit below.

What about winning the NBA finals? This remains a longshot for underdog teams, but the probability improves from practically 0% to 0.02% in a single series for an 8-seed (assuming a constant series win probability across four series). Over 20 years, the probability of winning a finals increases from 0.08% to 0.49%. Still *very* unlikely, but a significant improvement. See the second exhibit below.



3pt volume (% of total shots)	Probability of winning series	Prob of winning at least 1 series in 20	Prob of winning NBA finals	Prob of winning NBA finals in 20 attempts
30%	8.00%	81.13%	0.00%	0.08%
60%	12.50%	93.08%	0.02%	0.49%

What about real-life winners?

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What do we observe when we compare real-life lower-seeded teams that actually pull off upsets versus those that lose? Across the last decade, lower seeded teams that beat higher seeded teams took a higher share of three point shots, shooting them at roughly a 4 to 6 percent higher rate on average. Winning underdogs (6-8 seeds) also shot far more efficiently than expected, averaging around 39 percent from three in first-round wins.

For example, the 8-seed Miami Heat defeated the Milwaukee Bucks in the playoffs of the 2022-2023 season–the only 8 seed to beat a 1 seed over the past ten years. The Heat slightly reduced their three-point volume compared to the regular season but shot an incredible 45 percent from three in the series or 10 percent higher than their regular season average, which was a major driver of the upset (with an additional caveat that the fact that Giannis Antetokounmpo missed time with a back injury). This underscores that shooting more 3's alone isn't enough for an underdog to win, but also the luck of shooting efficiently.

<u>Seeds 6-8 in the 1st round of the NBA Playoffs 2016 - 2025:</u>

seed	series_result	Series	reg_3pa_rate	series_3pa_rate	delta_3pa_rate	reg_3p_pct	series_3p_pct	delta_3p_pct
6	lost	16	37.2%	36.0%	-1.2%	36.2%	34.1%	-2.1%
6	won	4	40.8%	40.7%	-0.2%	37.5%	34.9%	-2.6%
7	lost	17	34.9%	36.2%	1.3%	35.8%	34.3%	-1.5%
7	won	3	40.9%	42.1%	1.2%	35.1%	37.2%	2.1%
8	lost	19	34.0%	34.5%	0.5%	36.1%	33.9%	-2.1%
8	Won	1	40.8%	38.1%	-2.7%	34.4%	45.3%	10.9%

However, one thing to note is that first round match-ups in recent years have actually been somewhat even, particularly in the Western Conference. For example, last year we saw the 7 seed Warriors as a 64% favorite (implied by betting odds) against the 2 seeded Rockets.

Perhaps a better way to view these results is by using sports betting odds versus seeding. When looking at all playoffs series in the last 10 years we see some interesting results. Below we break out winners vs losers based on pre-series betting odds (15 series each year, 10 years, 150 data points). We see there is actually no large difference in 3-point attempt rate.

Infact, big underdogs - ones who have an expected win percentage of below 17% - actually shoot less threes. However, these underdogs who have overcome those odds shot 39% from 3 versus the favorites who shot 32%. The favorites on average shoot 5% worse from 3 versus in the regular season and the underdogs shoot 2% better! This can be seen in the yellow box below!

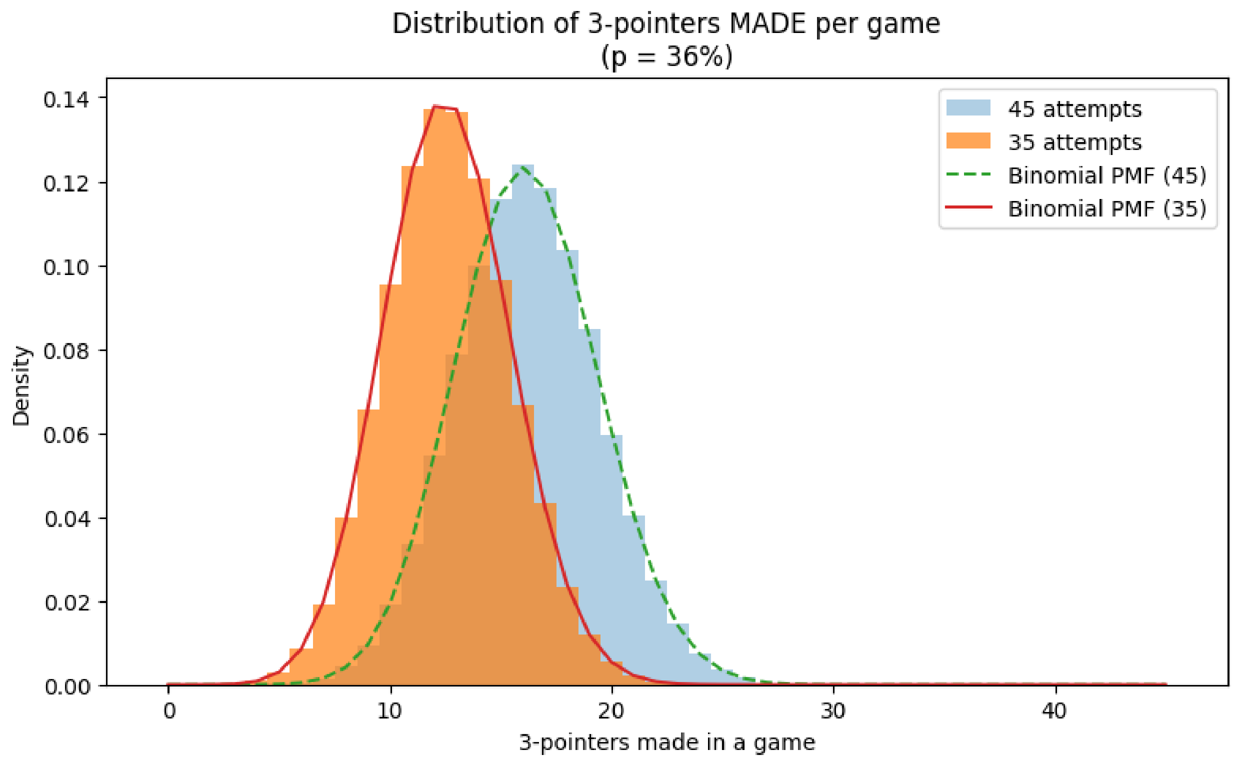
All Playoff Outcomes by Betting Odds 2016-2025													
Underdog Win%	Odds_Bucket	Role	Outcome	REG_3PA_rate	SER_3PA_rate	REG_FG3_PCT	SER_FG3_PCT	REG_2PA_rate	SER_2PA_rate	REG_FG2_PCT	SER_FG2_PCT	Underdog Win %	Count
33-50%	-200 to -100	Fav	Lost Series	39.0%	39.3%	37.1%	35.0%	61.0%	60.7%	54.2%	51.8%	53.3%	24
33-50%	-200 to -100	Und	Won Series	38.8%	38.2%	37.1%	35.7%	61.2%	61.8%	54.7%	53.1%	53.3%	24
33-50%	-200 to -100	Fav	Won Series	39.2%	40.4%	36.5%	36.6%	60.8%	59.6%	54.0%	52.6%	53.3%	21
33-50%	-200 to -100	Und	Lost Series	37.5%	36.5%	36.5%	34.1%	62.5%	63.5%	52.8%	49.7%	53.3%	21
17-33%	-500 to -200	Fav	Lost Series	36.9%	35.9%	37.2%	35.2%	63.1%	64.1%	54.0%	50.8%	20.4%	10
17-33%	-500 to -200	Und	Won Series	37.1%	38.8%	36.3%	37.2%	62.9%	61.2%	53.9%	52.6%	20.4%	10
17-33%	-500 to -200	Fav	Won Series	38.9%	39.5%	37.2%	35.6%	61.1%	60.5%	54.0%	53.2%	20.4%	39
17-33%	-500 to -200	Und	Lost Series	37.2%	38.3%	36.2%	34.3%	62.8%	61.7%	53.1%	50.0%	20.4%	39
<17%	< -500	Fav	Lost Series	42.3%	39.7%	37.1%	32.3%	57.7%	60.3%	54.8%	52.2%	12.5%	7
<17%	< -500	Und	Won Series	37.6%	37.6%	36.2%	38.9%	62.4%	62.4%	53.1%	51.3%	12.5%	7
<17%	< -500	Fav	Won Series	39.2%	39.1%	37.6%	38.1%	60.8%	60.9%	54.8%	55.3%	12.5%	49
<17%	< -500	Und	Lost Series	35.6%	36.5%	36.2%	33.7%	64.4%	63.5%	52.5%	49.4%	12.5%	49

The math behind these results

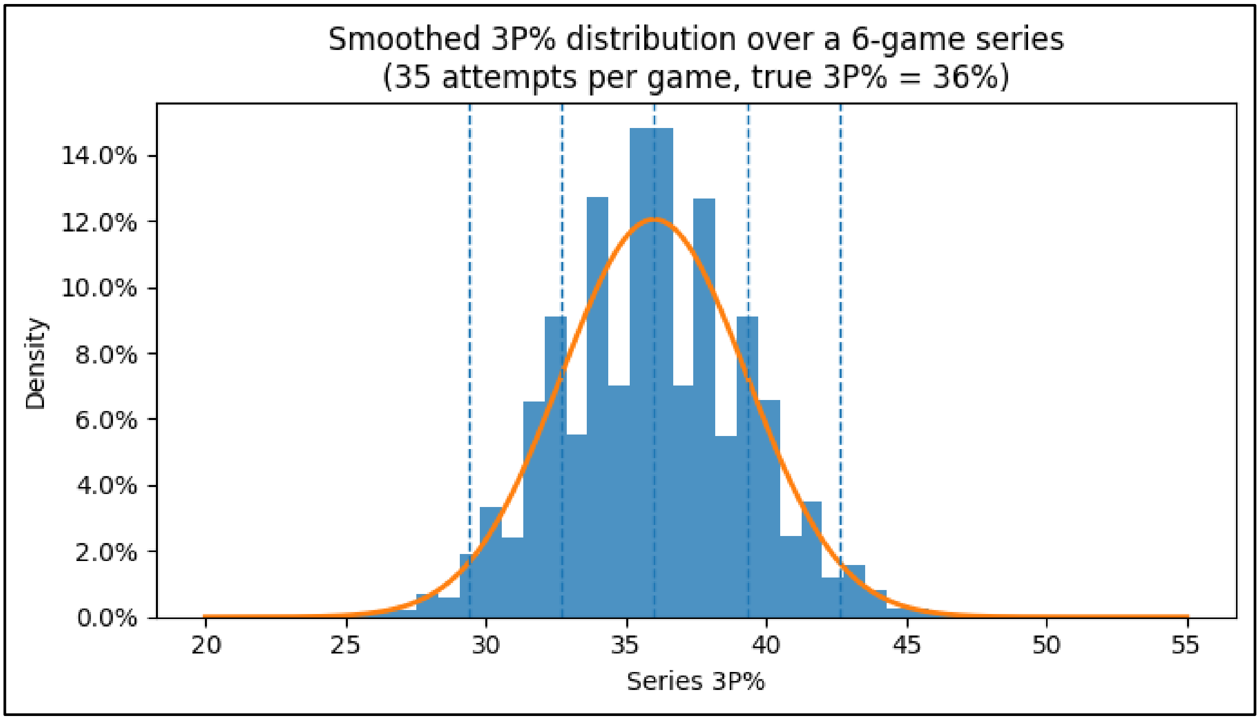
Are the winners doing something different or is it just a feature of variance? Let's take the following Monte Carlo (n=1,000,000) simulation:

- Suppose a team's true mean is to convert 36% of three-point shots attempted
- The underdog shoots 35 three-point shots and 45 two-point shots in a game
- Shooting pattern follows a binomial distribution

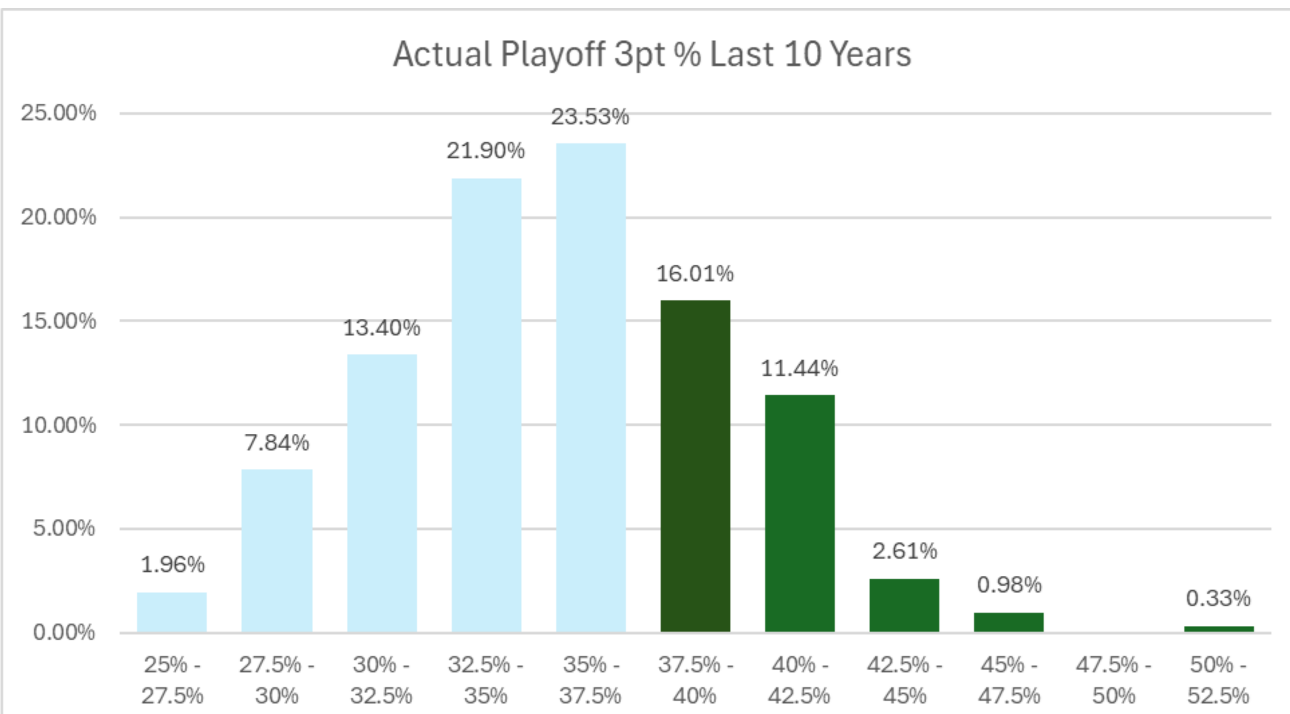
Increasing three-point volume from 35 to 45 attempts per game (assuming a fixed 90 total attempts) systematically changes the statistical properties of a team's shooting outcomes. Because made three-pointers follow a binomial process with parameters n (shot volume) and p (true accuracy), the expected number of makes increases proportionally with n, rising from 12.6 makes per game at 35 attempts to 16.2 makes at 45 attempts for a team that for a team that makes 36% of its three-point attempts. The dispersion of outcomes also increases: the standard deviation grows from 2.84 to 3.18 when volume rises from 35 to 45 shots. This reflects the fundamental property that binomial variance, given by $n \times p \times (1 - p)$, increases with shot volume. As a result, higher three-point volume not only raises expected scoring but also produces a wider distribution of possible game-level outcomes.



If a team is one standard deviation to the right of its true shooting ability, it will shoot above roughly 39.4 percent from three about 16 percent of the time in a 6-game series. That lines up nicely with the empirical rate at which lower-seeded teams have delivered elite shooting performances in upsets: 8 out of 60, or about 13.3 percent. The Miami Heat's shooting in its win over Milwaukee is a good example of a true +2 standard deviation outcome: shooting better than 42.5 percent from three over a series is only about a 2.5 percent event in the model, which closely matches the empirical frequency of extreme outlier performances (1 out of 60, or 1.66 percent). In other words, what we observe in real playoff upset shooting percentages aligns pretty nicely with what simple variance math would predict.



In real life, teams shoot above 40% from three in the playoffs over a series around 15% of the time. This matches our simulated results fairly nicely (over 39.4%, 16% of the time).



Data: last 10 years of playoff series data, 15 series per year, data on 300 teams shooting percentages.

Appendix: Additional Exhibits

Exhibit A: Regular Season versus Post Season Shooting Percentage (Data from NBA.com)

seed	reg_3pa_rate	series_3pa_rate	delta_3pa_rate	reg_3p_pct	series_3p_pct	delta_3p_pct	reg_2pa_rate	series_2pa_rate	delta_2pa_rate	reg_2p_pct	series_2p_pct	delta_2p_pct
1	40.3%	40.9%	0.6%	37.3%	37.9%	0.6%	59.7%	59.1%	-0.6%	55.1%	55.0%	0.0%
2	37.6%	37.0%	-0.5%	37.1%	37.3%	0.2%	62.4%	63.0%	0.5%	53.4%	52.3%	-1.1%
3	37.2%	38.0%	0.9%	36.6%	36.5%	-0.1%	62.8%	62.0%	-0.9%	54.0%	52.3%	-1.7%
4	37.9%	38.6%	0.7%	36.8%	35.6%	-1.3%	62.1%	61.4%	-0.7%	53.5%	51.4%	-2.1%
5	37.1%	38.2%	1.1%	36.4%	33.5%	-2.9%	62.9%	61.8%	-1.1%	53.0%	50.9%	-2.1%
6	37.9%	37.0%	-1.0%	36.5%	34.3%	-2.2%	62.1%	63.0%	1.0%	53.3%	50.7%	-2.6%
7	35.8%	37.1%	1.3%	35.7%	34.7%	-1.0%	64.2%	62.9%	-1.3%	52.1%	50.0%	-2.1%
8	34.3%	34.7%	0.4%	36.0%	34.5%	-1.5%	65.7%	65.3%	-0.4%	52.1%	49.7%	-2.4%